

Kapture Finance AI Collections Voicebot

High-Level Design (HLD)

AI Delivery Intern – Take-Home Assignment

Agent: Maya – Outbound Collections Voicebot

Platform: Vapi

Version: 1.0

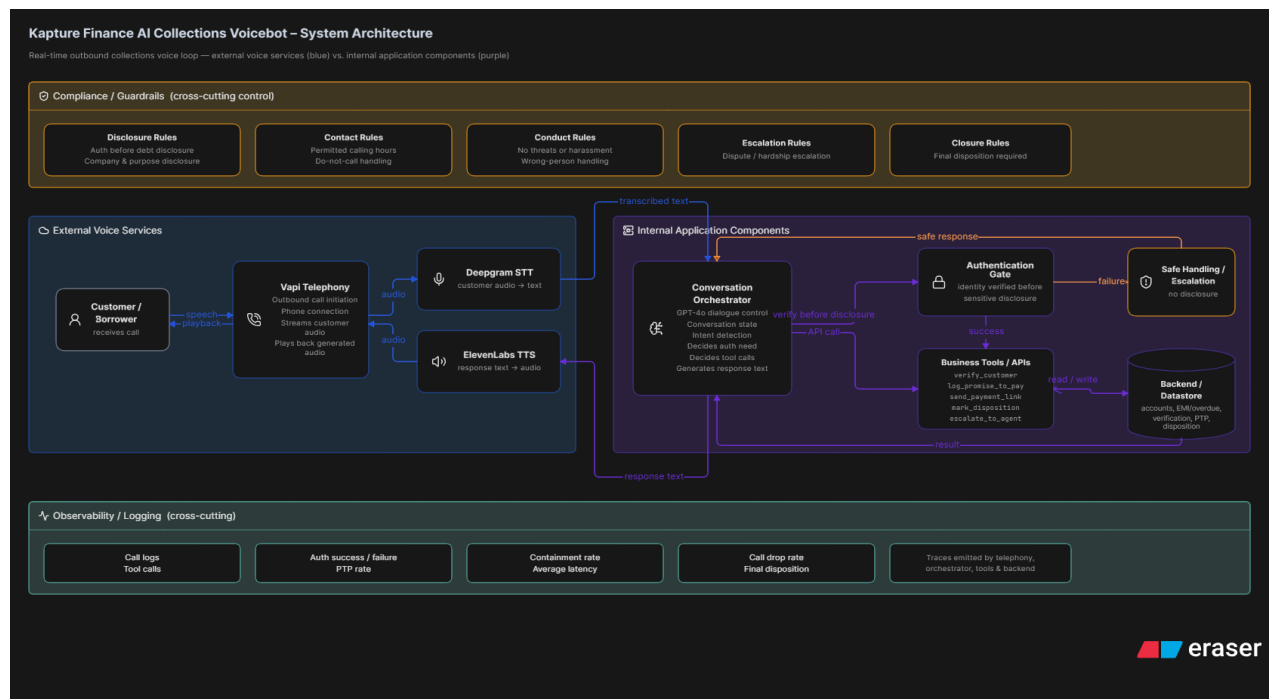
Date: August 2026

1. Executive Summary

Maya is an outbound AI voice collections assistant designed for Kapture Finance to handle routine overdue EMI conversations safely and consistently. The system uses telephony, speech-to-text, an LLM-based conversation orchestrator, text-to-speech, business tools, and backend services to authenticate customers, disclose account information only after successful verification, understand repayment intent, record promises to pay, provide payment links, handle disputes and edge cases, and record a final call disposition.

The design uses explicit conversation states so that authentication is enforced before any sensitive loan information is disclosed. Business actions such as verification, promise-to-pay logging, payment-link generation, escalation, and disposition recording are performed through controlled tools rather than relying only on prompt instructions.

2. Architecture & Pipeline



The voicebot follows a layered pipeline. An outbound call enters through the telephony layer and is converted to text through speech-to-text. The conversation orchestrator manages the state machine, determines customer intent, and selects the appropriate business tool. Authentication is enforced

before account or debt information is disclosed. Tool calls interact with backend services and return structured results to the orchestrator. The response is converted back to speech using the text-to-speech layer. Every completed interaction produces a final disposition for monitoring and reporting.

Latency Budget

Component	Target
Telephony	<300 ms
Speech-to-Text	<500 ms
LLM / Orchestration	<1,000 ms
Business Tool/API	<500 ms
Text-to-Speech	<500 ms
Target total response latency	<2.5 sec

3. Conversation Flow & State Machine

The collections voicebot follows an explicit state-based conversation flow. Sensitive account information is only disclosed after successful customer authentication.

Conversation States

CALL_INIT → INTRODUCTION → CONSENT → AUTHENTICATION → ACCOUNT_DISCLOSURE → INTENT_DETECTION → ACTION → DISPOSITION → CALL_END

State Rules

State	Purpose	Transition Rule
CALL_INIT	Start outbound call	Move to Introduction
INTRODUCTION	Introduce Kapture Finance and call purpose	Customer engages → Consent
CONSENT	Confirm customer is willing to continue	Consent given → Authentication
AUTHENTICATION	Verify customer identity	Success → Account Disclosure; Failure → Retry/End
ACCOUNT_DISCLOSURE	Provide overdue account information	Only accessible after successful authentication
INTENT_DETECTION	Understand customer's response	Route to appropriate intent

State	Purpose	Transition Rule
ACTION	Execute payment, escalation or opt-out action	Action completed → Disposition
DISPOSITION	Record final outcome	Completed → Call End
CALL_END	Close the interaction	End

Critical Security Rule

Authentication is **state-enforced**. The agent must never disclose the overdue amount, loan details, or other sensitive account information before AUTHENTICATION = SUCCESS.

4. Intents & Entities

The agent identifies the customer's intent and extracts the relevant information required to resolve the call.

Intent	Agent Action	Entities
Will Pay	Record Promise to Pay	Payment date, amount
Cannot Pay / Hardship	Understand situation and escalate if required	Reason, callback preference
Already Paid	Acknowledge and verify/escalate	Payment date, reference
Dispute	Do not argue; escalate	Dispute reason, amount
Do Not Call	Record opt-out and end call	Opt-out status
Wrong Person	Do not disclose debt; end call	Relationship
Callback Request	Arrange appropriate callback	Date, time
Hostile Customer	De-escalate and escalate/end	—

5. Tools / API Calls

The Vapi assistant uses controlled tools for business actions rather than relying only on LLM instructions.

5.1 verify_customer

Purpose: Authenticate the customer before any account information is disclosed.

Inputs:

- customer_id
- verification_code

Output:

- Verification status
- Customer verification result

Used: During the Authentication state.

5.2 log_promise_to_pay

Purpose: Record a customer's payment commitment.

Inputs:

- customer_id
- promise_date
- promise_amount

Output:

- PTP recorded successfully

Used: When the customer agrees to make a payment.

5.3 send_payment_link

Purpose: Send a secure payment link to a verified customer.

Inputs:

- customer_id

Output:

- Payment link status

Used: After successful authentication when payment-link delivery is required.

5.4 mark_disposition

Purpose: Record the final outcome of the call.

Inputs:

- customer_id
- disposition

Output:

- Disposition recorded successfully

Used: Before completing the call.

5.5 end_call_tool

Purpose: Safely terminate the conversation after the required outcome has been recorded.

Used: At the end of a completed interaction.

6. Authentication & Data Safety

The agent follows a verification-before-disclosure policy.

1. The agent introduces itself and explains the purpose of the call.
2. The customer provides the required authentication information.
3. The verify_customer tool validates the information.
4. Only after successful verification can the agent disclose overdue EMI or account information.
5. If verification fails, the agent does not reveal sensitive information.
6. If another person answers the call, the agent does not disclose the customer's debt information.

Authentication is therefore enforced as a conversation state and supported by a dedicated verification tool.

7. Guardrails & Compliance

The voicebot follows these mandatory guardrails:

- Clearly identify Kapture Finance and the purpose of the call.
- Never disclose debt information before successful authentication.
- Never threaten, harass or pressure the customer.
- Respect Do-Not-Call requests.
- Do not reveal sensitive information to a wrong person.
- Do not invent account, payment or loan information.
- Handle disputes without arguing with the customer.
- Escalate situations requiring human assistance.
- Respect permitted calling hours.
- Record the final disposition of every completed interaction.

8. Edge Cases

The system handles the following cases:

Already Paid

The agent acknowledges the customer's statement, avoids arguing, and routes the case for verification or human follow-up.

Dispute

The agent records that the customer disputes the amount and escalates the case rather than attempting to force payment.

Do Not Call

The agent acknowledges the request, records the opt-out disposition and ends the call.

Wrong Person

The agent does not reveal debt information and ends the interaction without disclosing the purpose beyond the permitted introduction.

No Input / Timeout

The agent provides a concise retry prompt. If there is still no response, the call is ended and the appropriate disposition is recorded.

Hostile Customer

The agent remains polite, avoids confrontation and escalates or ends the call when necessary.

Language Switch

The agent can support a switch between English and Hindi while maintaining the same authentication and compliance rules.

9. Escalation & Disposition

Human escalation is used for cases that cannot safely or appropriately be resolved by the automated agent, including disputes, hardship situations, complex account issues and customer requests for human assistance.

Every completed call must have a final disposition such as:

- PTP_COMMITTED
- ALREADY_PAID
- DISPUTE
- HARDSHIP
- DO_NOT_CALL
- WRONG_PERSON
- CALLBACK_REQUESTED
- ESCALATED

The mark_disposition tool records the final outcome before the call is completed.

10. Observability & Metrics

The system records operational and business metrics to identify failures and improve the voicebot.

Metric	Purpose
PTP Rate	Measures successful payment commitments
Containment Rate	Measures calls resolved without human intervention

Metric	Purpose
Authentication Success Rate	Measures verification effectiveness
Average Latency	Measures conversational responsiveness
Tool Failure Rate	Identifies backend/tool problems
Call Drop Rate	Identifies disconnected or failed calls
Escalation Rate	Measures cases requiring human support
Dispute Rate	Measures disputed accounts

Important events such as authentication results, tool calls, tool failures, state transitions and final dispositions should be logged for debugging and monitoring.

11. Assumptions & Design Choices

- Vapi is used as the voice-agent orchestration platform.
- The assistant uses speech-to-text, an LLM for conversation orchestration, and text-to-speech for natural voice interaction.
- Customer and loan information is assumed to be available through backend APIs or mocked endpoints.
- Authentication is mandatory before any sensitive account information is disclosed.
- Payment actions are performed through controlled tools rather than direct LLM-generated actions.
- The system is designed for routine collections interactions while complex cases are escalated to human agents.
- The prototype uses mocked business APIs where production integrations are unavailable.

Model and Voice Selection

The Vapi assistant is configured with deliberately selected speech, language-model and voice components to balance conversational quality, latency and cost. The final implementation prioritizes reliable tool calling and clear speech over unnecessary model complexity.

12. Testing Strategy

The voicebot should be tested against both successful and failure scenarios.

Test Case	Expected Result
Valid authentication	Account information can be disclosed
Invalid authentication	Account information remains hidden
Customer promises to pay	PTP is recorded

Test Case	Expected Result
Customer already paid	Case is handled without forcing payment
Customer disputes amount	Case is escalated
Do-not-call request	Opt-out is recorded and call ends
Wrong person answers	No sensitive information is disclosed
Customer requests human	Human escalation is initiated
Tool failure	Agent handles failure safely
No response	Retry followed by appropriate call termination

The prototype demonstrations validate the primary PTP flow and call-completion behavior.

13. Conclusion

The proposed Kapture Finance collections voicebot combines voice interaction, explicit conversation states, authentication controls, business tools and compliance guardrails to automate routine overdue-EMI conversations safely.

The architecture separates conversational reasoning from sensitive business actions through controlled tools. State-based authentication prevents premature disclosure of customer information, while escalation and disposition handling provide safe paths for disputes, hardship cases and other exceptions.

The design can be extended with production payment APIs, CRM integration, multilingual support, human-agent transfer and automated evaluation at scale.